

SAFA

SMASH Afterburner Femtoscopy Analysis

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1. Configuration

`main.yaml` — Run configuration

- ▶ Input and output paths
- ▶ Number of events and subevents
- ▶ Number of threads
- ▶ Enable/disable HBT
- ▶ Path to the HBT configuration

`hbt.yaml` — HBT analysis configuration

- ▶ Analysis products / particle-pair channels
- ▶ Particle acceptance cuts
- ▶ k_T and m_T slicing
- ▶ Histogram binning
- ▶ Origin selections
- ▶ Fit estimator(s)

`main.yaml` → `hbt.yaml`



Required channels



Required particle species

main.yaml

```
events_path: "../.."
output_path: "../output"
number_of_events: 1
number_of_subevents: 2000
threads: 1
hbt_enabled: true
hbt_config_path: "hbt.yaml"
```

hbt.yaml — Channels & Particle Acceptance

```
hbt_enabled_channels: >-
  pi_plus_pi_plus+pi_minus_pi_minus,
  p_p+p_bar_p_bar,
  pi_plus_p+pi_minus_p_bar,
  k_plus_p+k_minus_p_bar,
  k_plus_k_plus+k_minus_k_minus

hbt_particle_acceptance:
  # Allowed values:
  # - rapidity: apply the longitudinal cut to |y|
  # - pseudorapidity: apply the longitudinal cut to |eta|
  longitudinal_variable: "pseudorapidity"

groups:
  pions:
    longitudinal_abs_max: 0.8
    pt_min_gev: 0.14
    pt_max_gev: 4.0

  kaons:
    longitudinal_abs_max: 0.8
    pt_min_gev: 0.4
    pt_max_gev: 1.4

  nucleons:
    longitudinal_abs_max: 0.8
    pt_min_gev: 0.5
    pt_max_gev: 4.05

  sigmas:
    longitudinal_abs_max: 0.8
    pt_min_gev: 1.0
    pt_max_gev: 10000.0

  lambdas:
    longitudinal_abs_max: 0.8
    pt_min_gev: 0.3
    pt_max_gev: 10000.0
```

hbt.yaml — Pair Slicing & Histograms

```
# Pair-level slicing. Each enabled axis requires at least two consecutive
# bin edges in GeV. Disabled axes may retain validated production binning.
# Slices use half-open intervals [edge_i, edge_(i+1)).
hbt_pair_slicing:
  kt:
    enabled: false

  mt:
    enabled: true
    bin_edges_gev:
      - 0.50
      - 0.70
      - 0.90
      - 1.10
      - 1.30
      - 1.50
      - 1.70
      - 1.90
      - 2.10
      - 2.30
      - 2.50
      - 2.70

# Raw source histograms. These bins are independent of kT/mT slicing.
# OSL components and radial source radii are histogrammed from zero.
# Relative times retain their sign and therefore span zero.
hbt_histograms:
  osl:
    nbins: 3000
    min_fm: 0.0
    max_fm: 300.0

  radial:
    nbins: 3000
    min_fm: 0.0
    max_fm: 300.0

  delta_t:
    nbins: 6000
    min_fm_c: -300.0
    max_fm_c: 300.0
```

hbt.yaml — Origin Selection & Fit Estimator

```
# Allowed values:
# - primordial
# - primordial_rescattering
# - primordial_rescattering_decay
# - all: request all three nested origin selections in the same run
hbt_origin_mode: "all"

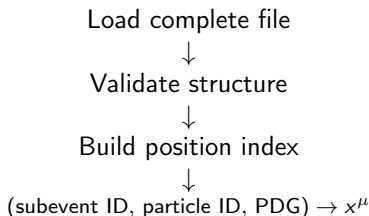
# Statistical estimator used for pure-Gaussian and mixed fits.
# Allowed values:
# - poisson
# - neyman
# - pearson
# - all: execute all three estimators independently
hbt_fit_estimator: "all"
```

2. Event Reading

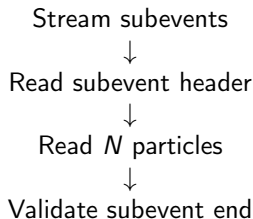
events_path/<event>/{Sampler, Afterburner}/particle_lists.oscar

Outer event

Sampler



Afterburner



Event processing

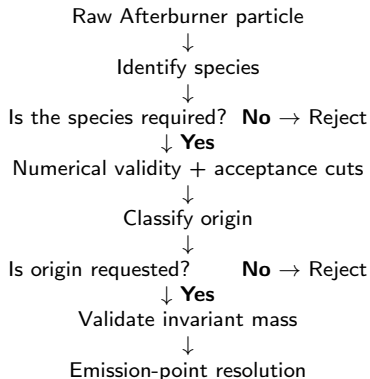
Matching requirement: The subevent ID is essential for matching: particle ID and PDG alone are not sufficient to uniquely associate an Afterburner particle with its Sampler emission point.

Input consistency: Sampler must contain the configured number of subevents, and Afterburner must contain exactly that number — neither fewer nor more.

Input failure: An input failure at this level causes the **whole outer event to be discarded**.

3. Particle Preparation and Selection

Selection flow



Configured acceptance cuts

Species	Longitudinal	p_T [GeV]
Pions	$ \eta < 0.8$	$0.14 < p_T < 4.0$
Kaons	$ \eta < 0.8$	$0.40 < p_T < 1.4$
Nucleons	$ \eta < 0.8$	$0.50 < p_T < 4.05$
Sigmas	$ \eta < 0.8$	$1.00 < p_T < 10000$
Lambdas	$ \eta < 0.8$	$0.30 < p_T < 10000$

The acceptance cuts are open:

$$|\eta| < \eta_{\max}, \quad p_{T,\min} < p_T < p_{T,\max}.$$

Important distinction

Physical cut rejection and numerical rejection are treated separately.

Examples of numerical rejection include non-finite momentum, $E \leq 0$, or an invalid pseudorapidity calculation.

3. Emission Point Resolution

Accepted candidate

Primordial and $n_{\text{coll}} = 0$

Otherwise

Sampler position is **mandatory**



Match using

(subevent ID, particle ID, PDG)



Use Sampler emission point

Is time_last_coll usable?

Yes



Propagate
backwards
to last interaction

No



Use final
Afterburner
position

Sampler

Propagation

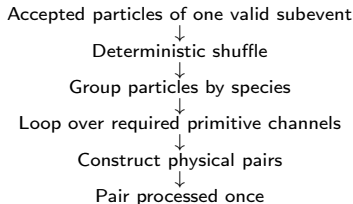
Afterburner

Mandatory Sampler match

For a primordial particle with $n_{\text{coll}} = 0$, the Sampler match is mandatory. If the required match cannot be found, SAFA raises a recoverable subevent failure and the **subevent is discarded**.

4. Accepted Particles and Pair Construction

Pair-construction flow



Why shuffle?

SMASH input ordering may contain non-physical correlations. The shuffle removes ordering effects while preserving reproducibility.

Seed:
(outer event number, subevent ID)

Identical species

Examples: $p-p$, $\pi^+-\pi^+$

$$i < j \implies N_{\text{pairs}} = \frac{N(N-1)}{2}$$

Each unordered physical pair is produced **exactly once**.

Different species

Example: K^-p

$$A \times B \implies N_{\text{pairs}} = N_A N_B$$

The full Cartesian product is generated, preserving the predefined **species-A/species-B** roles of the channel.

Processing scope

- ▶ Only primitive channels required by the configured analysis products are processed.
- ▶ Pairs are not stored in memory: each pair is passed directly to the next processing stage.

4. Pair Reuse: Origins and Analysis Products

Key design principle: Each physical pair is constructed and processed only once for its primitive channel.

Physical pair $\longrightarrow$ Primitive channel $\longrightarrow$ Observables computed once

Origin fan-out

The same processed pair can contribute to more than one compatible origin selection.

With `origin_mode: all`, the selections are nested. For example,

$$P-P \longrightarrow P, PR, PRD.$$

No pair is reconstructed for the different origin routes.

Product fan-out

A primitive channel may belong to several configured final analysis products.

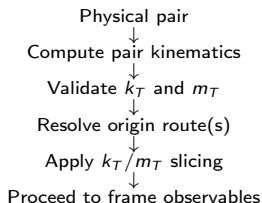
$$\begin{array}{lcl} & & \text{product 1} \\ \text{primitive channel} & \longrightarrow & \text{product 2} \\ & & \vdots \end{array}$$

The already-computed pair contribution is routed to every product that contains that primitive channel.

No duplicated pair calculation: Overlap between origin selections or analysis products only changes **where the pair contribution is accumulated**; it does not trigger a new pair construction or a new pair-observable calculation.

5. Pair Processing

Pair-level flow



Numerical validation

Non-finite k_T or m_T causes a **numerical rejection**. The rejected pair is recorded together with its event, subevent, primitive channel and rejection reason.

Pair kinematics

$$\vec{P} = \vec{p}_a + \vec{p}_b,$$

$$k_T = \frac{1}{2} \sqrt{(p_{x,a} + p_{x,b})^2 + (p_{y,a} + p_{y,b})^2},$$

$$m_T = \sqrt{\left(\frac{m_a + m_b}{2}\right)^2 + k_T^2}.$$

k_T/m_T slicing

If slicing is enabled, SAFA determines the slice **once** from the already-computed pair kinematics.

$$x_i \leq x < x_{i+1}.$$

If both axes are enabled, the pair receives a combined (k_T, m_T) slice.

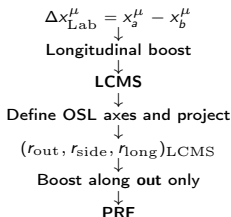
Outside the configured slice

A numerically valid pair outside the configured k_T/m_T range is **not** a numerical rejection. It simply does not continue to the frame-observable / histogram stage.

Pair kinematics are calculated once and reused downstream

6. Frame Transformations: Lab $\rightarrow$ LCMS $\rightarrow$ PRF

Transformation flow



Exact $k_T = 0$ case

When $k_T = 0$, the out direction is defined from the transverse relative momentum $\vec{q}_T = \vec{p}_{T,a} - \vec{p}_{T,b}$.

Lab $\rightarrow$ LCMS

$$\beta_{\text{LCMS}} = \frac{P_z}{E}, \quad \gamma_{\text{LCMS}} = \frac{1}{\sqrt{1 - \beta_{\text{LCMS}}^2}}.$$

$$\Delta t_{\text{LCMS}} = \gamma_{\text{LCMS}}(\Delta t_{\text{Lab}} - \beta_{\text{LCMS}} \Delta z_{\text{Lab}}),$$

$$\Delta z_{\text{LCMS}} = \gamma_{\text{LCMS}}(\Delta z_{\text{Lab}} - \beta_{\text{LCMS}} \Delta t_{\text{Lab}}).$$

The transverse coordinates are unchanged.
OSL basis in the LCMS

$$\hat{e}_{\text{out}} = \frac{(K_x, K_y)}{k_T}, \quad \hat{e}_{\text{side}} = (-e_{\text{out},y}, e_{\text{out},x}),$$

with the long direction along the beam z axis.
LCMS $\rightarrow$ PRF

$$\beta_{\text{out}} = \frac{2k_T \gamma_{\text{LCMS}}}{E}, \quad \gamma_{\text{out}} = \frac{1}{\sqrt{1 - \beta_{\text{out}}^2}}.$$

$$\Delta t_{\text{PRF}} = \gamma_{\text{out}}(\Delta t_{\text{LCMS}} - \beta_{\text{out}} r_{\text{out}}^{\text{LCMS}}),$$

$$r_{\text{out}}^{\text{PRF}} = \gamma_{\text{out}}(r_{\text{out}}^{\text{LCMS}} - \beta_{\text{out}} \Delta t_{\text{LCMS}}).$$

$$r_{\text{side}}^{\text{PRF}} = r_{\text{side}}^{\text{LCMS}}, \quad r_{\text{long}}^{\text{PRF}} = r_{\text{long}}^{\text{LCMS}}$$

6. HBT Pair Observables

All frame observables are calculated **once for the physical pair** and then reused by the histogram stage.

Emission-time differences

$$\Delta t_{\text{Lab}}, \quad \Delta t_{\text{LCMS}}, \quad \Delta t_{\text{PRF}}$$

Radial separations

$$r_{\text{LCMS}} = \sqrt{(r_{\text{out}}^{\text{LCMS}})^2 + r_{\text{side}}^2 + r_{\text{long}}^2},$$

$$r_{\text{PRF}} = \sqrt{(r_{\text{out}}^{\text{PRF}})^2 + r_{\text{side}}^2 + r_{\text{long}}^2}.$$

Spatial components

$$r_{\text{out}}^{\text{LCMS}}, \quad r_{\text{out}}^{\text{PRF}},$$

$$r_{\text{side}}, \quad r_{\text{long}}$$

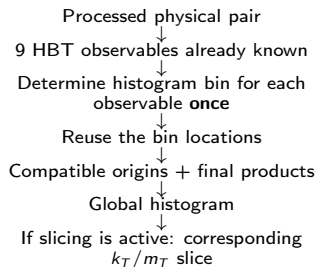
Numerical validity

After the transformations, every frame observable is checked for finiteness. A non-finite value causes a **numerical pair rejection**.

Frame observables computed once → reused downstream

7. Histogram Accumulation

Accumulation flow



Raw counts only

At this stage SAFA stores **raw integer counts**. No normalization, fitting, or statistical analysis is performed here.

Nine raw histograms per destination
OSL components — absolute values

$$|r_{\text{out}}^{\text{LCMS}}|, \quad |r_{\text{out}}^{\text{PRF}}|, \quad |r_{\text{side}}|, \quad |r_{\text{long}}|$$

Radial distributions

$$r_{\text{LCMS}}, \quad r_{\text{PRF}}$$

Relative-time distributions — signed

$$\Delta t_{\text{Lab}}, \quad \Delta t_{\text{LCMS}}, \quad \Delta t_{\text{PRF}}$$

Global and sliced histograms

With slicing active, an accepted pair contributes to the **global histogram** and to its **single k_T/m_T slice**.

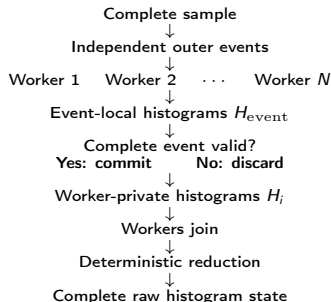
Histogram boundaries

Values outside the configured range are counted explicitly as **underflow** or **overflow**; in-range values increment the corresponding bin.

Bin classification once → reuse across origins and products

8. Full-Sample Processing and Deterministic Reduction

Full-sample flow



Transactional event processing

An event contributes only if the **complete outer event** is valid. An event-level input failure discards all histogram contributions produced by that event.

Commit rule

Each outer event first fills a temporary state:

$$H_{\text{event}}.$$

Only after successful completion is it committed:

$$H_{\text{worker}} \leftarrow H_{\text{worker}} + H_{\text{event}}.$$

Subevent vs. event failure

A recoverable **subevent** failure skips that subevent while other valid subevents of the same outer event may remain usable. An **event-level input failure** discards the whole outer event transaction.

Parallel execution and reduction

Each worker owns a private histogram state H_i ; histogram bins are never updated concurrently by several workers. After all workers join,

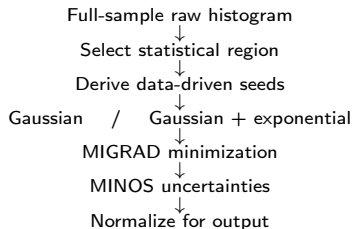
$$H_{\text{total}} = H_1 + H_2 + \dots + H_N.$$

Bins, underflows, and overflows are reduced **bin by bin** using integer counters.

Parallel scheduling may change; the scientific result does not.

9. Statistical Analysis: From Raw Counts to Fits

Post-sample analysis flow



Raw counts drive the fits

The statistical objectives operate directly on the **raw integer histogram counts**. Normalization is performed only **after** the fits and Δt statistics are complete.

Shape-region selection

For OSL and radial histograms, the selected region starts at bin zero and ends immediately before the **first empty bin to the right of the first modal plateau**.

Leading empty bins are retained; later disconnected islands are not reincorporated.

Pure Gaussian fit region

In the current production path, the pure Gaussian uses the **complete selected shape region**. No external 5% or 10% Gaussian-core threshold is applied to the fit.

Estimator independence

If `hbt_fit_estimator: all`, Poisson, Neyman and Pearson are run as **three independent fits**. Their minima, seeds anchored to fitted results, and uncertainties are not shared.

9. Fit Models and Statistical Estimators

Pure Gaussian

OSL:

$$G(x; R) \propto \exp\left(-\frac{x^2}{4R^2}\right)$$

Radial:

$$G(r; R) \propto r^2 \exp\left(-\frac{r^2}{4R^2}\right)$$

Parameters:

$$R, \quad A_G > 0$$

The Gaussian amplitude is free and positive.

Gaussian + exponential

$$P = f_{\text{core}} G(R_{\text{core}}) + (1 - f_{\text{core}}) E(R_{\text{tail}})$$

OSL: $E(x; R_{\text{tail}}) \propto e^{-x/R_{\text{tail}}}$

Radial: $E(r; R_{\text{tail}}) \propto r^2 e^{-r/R_{\text{tail}}}$

The two components are normalized independently over the selected fit region before mixing.

Implemented estimators

Poisson deviance

$$D = 2 \sum_i \left[\mu_i - n_i + n_i \ln\left(\frac{n_i}{\mu_i}\right) \right]$$

with the $n_i = 0$ contribution retained as $2\mu_i$.
Neyman χ^2

$$\chi_N^2 = \sum_{n_i > 0} \frac{(n_i - \mu_i)^2}{n_i}$$

Zero-count observed bins are omitted.
Pearson χ^2

$$\chi_P^2 = \sum_i \frac{(n_i - \mu_i)^2}{\mu_i}$$

Zero-count observed bins remain in the sum.

Independent fits

Poisson, Neyman and Pearson each execute their own Gaussian and mixed-model minimizations.

9. Gaussian Fit: Seed Generation

Seed 1: moments

From the selected raw histogram:

OSL

$$R_{\text{moment}} = \sqrt{\frac{\langle x^2 \rangle}{2}}$$

Radial

$$R_{\text{moment}} = \sqrt{\frac{\langle r^2 \rangle}{6}}$$

Seed 2: half maximum

A shape-constrained PAVA estimate is used only to identify the half-maximum width and obtain

$$R_{\text{HM}}.$$

The raw fit counts are never replaced or smoothed.

Two independent Gaussian starts

$$R_0 \in \{R_{\text{moment}}, R_{\text{HM}}\}$$

For each radius seed, SAFA derives the amplitude seed from the exact Gaussian integral over the selected region:

$$A_G^{(0)} = \frac{1}{\int_{\text{selected}} G(x; R_0) dx}$$

(with the corresponding radial integral for radial fits).

moment start $\rightarrow$ MIGRAD

half-maximum start $\rightarrow$ MIGRAD



Select the valid start with the lowest q

Then MINOS

After the Gaussian minimum is selected, MINOS is run for both the radius and the free amplitude.

9. Gaussian + Exponential: 36 Deterministic Seeds

Core-radius seeds

$$R_{\text{core}}^{(0)} \in \{R_G, 0.5R_{\text{HM}}, R_{\text{HM}}, 2R_{\text{HM}}\}$$

R_G is the Gaussian result from the **same estimator**: Poisson $\rightarrow$ Poisson, Neyman $\rightarrow$ Neyman, Pearson $\rightarrow$ Pearson.

Exponential moment seed

OSL:

$$R_{\text{tail,moment}} = \langle x \rangle$$

Radial:

$$R_{\text{tail,moment}} = \frac{\langle r \rangle}{3}$$

Tail starts:

$$R_{\text{tail}}^{(0)} = R_{\text{tail,moment}} \times \{0.5, 1, 2\}$$

Initial core fractions

$$f_{\text{core}}^{(0)} \in \{0.25, 0.50, 0.75\}$$

The deterministic grid therefore contains

$$4 \times 3 \times 3 = \boxed{36 \text{ MIGRAD starts per estimator}}$$

R_{core}	R_{tail} scale	f_{core}
4 choices	3 choices	3 choices

Why multi-start?

The mixed objective can contain several numerical minima. SAFA explores the same deterministic grid every time, then groups the converged endpoints into solution basins before deciding which physical solution to publish.

9. Selecting the Best Mixed-Fit Solution

1. Group converged endpoints into basins

Each valid start ends at

$$(\log R_{\text{core}}, \log R_{\text{tail}}, f_{\text{core}}).$$

Two endpoints are connected when

$$|\Delta \log R_{\text{core}}| \leq 0.01,$$

$$|\Delta \log R_{\text{tail}}| \leq 0.01, \quad |\Delta f_{\text{core}}| \leq 0.01.$$

2. Apply physical f_{core} cuts to the basin mean PRD:

$$0.1 < \langle f_{\text{core}} \rangle < 0.9$$

P and PR:

$$0.1 < \langle f_{\text{core}} \rangle < 0.99$$

3. Choose the winning basin

Basins are ranked by:

1. largest number of converged starts;
2. if tied, smallest q reached by the basin;
3. if still tied, lowest deterministic start index.

Selection principle

Robust convergence wins before minimum q .

4. Final polishing and errors

Members of the selected basin are ordered by terminal q . SAFA reruns a fresh MIGRAD + MINOS from the best endpoint.

If it fails, the next endpoint is tried inside the same selected basin. SAFA does not switch to another basin.

No condition such as

$$R_{\text{tail}} > R_{\text{core}}$$

is imposed.

9. Fit Quality Cuts and Parameter Errors

MIGRAD validity requirements

A candidate is rejected when, among other conditions:

- ▶ the objective evaluation fails;
- ▶ the MIGRAD call limit is reached;
- ▶ the EDM remains above the allowed Minuit level;
- ▶ $q_{\min}$ is unavailable / non-finite;
- ▶ the covariance is invalid;
- ▶ Minuit marks the minimum as invalid.

Current radial- m_T statistics threshold

The radial one-dimensional m_T quality-cut machinery exists, but the production threshold is currently

$$N_{\min} = 0.$$

It therefore does not veto any non-empty slice.

MINOS uncertainties

For all three estimators,

$$U_p = 1.$$

Thus MINOS searches for

$$\Delta(-2 \ln L) = 1 \quad (\text{Poisson}),$$

$$\Delta\chi^2 = 1 \quad (\text{Neyman / Pearson}).$$

Radii are optimized in $\log R$. MINOS first obtains the lower and upper limits in log-space; SAFA transforms both endpoints back to physical R -space:

$$R = R_{\text{fit}}^{+\sigma_+}_{-\sigma_-}.$$

The resulting radius errors are therefore naturally asymmetric.

f_{core} boundary requirement

MINOS is applied directly to f_{core} . The final interval must remain strictly inside $0 < f_{\text{core}} < 1$; touching or crossing either boundary rejects the solution.

9. Δt Statistics and Counting Errors

Signed Δt distributions

These histograms are **not fitted** with the Gaussian / mixed-model machinery. The selected region is bounded by the first empty bin on each side of the modal plateau. From the raw selected counts SAFA computes

$$\langle t \rangle,$$

$$\sigma_{\Delta t} = \sqrt{\langle t^2 \rangle - \langle t \rangle^2}.$$

For $N > 1$, the width uncertainty is

$$\delta \sigma_{\Delta t} = \frac{\sigma_{\Delta t}}{\sqrt{2(N-1)}}$$

If the variance is negative/non-finite, or if there is insufficient count for the uncertainty, SAFA reports the corresponding invalid/insufficient status rather than forcing a value.

Normalized histogram counting errors

Only after the statistical analysis, each selected histogram is normalized for presentation/output.

For bin i , with raw count n_i , total selected count N , and bin width Δx :

$$p_i = \frac{n_i}{N \Delta x},$$

$$\delta p_i = \frac{\sqrt{n_i}}{N \Delta x}$$

These are the counting errors written with the normalized distributions.

Two different uncertainty concepts

- ▶ Fit-parameter uncertainties: **MINOS**.
- ▶ Normalized-bin uncertainties: **Poisson counting error** $\sqrt{n_i}$ propagated through the normalization.

10. Output Organization

Production hierarchy

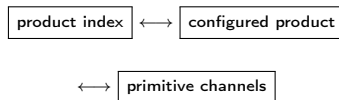
```
output_path/  
|-- product_catalog.csv  
|-- product_0/  
|   |-- global/  
|   |   |-- P/  
|   |   |-- PR/  
|   |   '-- PRD/  
|   '-- <configured-slice-token>/  
|       |-- P/  
|       |-- PR/  
|       '-- PRD/  
-- product_1/ ...
```

Inside each origin, results are organized by frame, observable family and observable, e.g.

```
LCMS/osl/r_out/  
LCMS/radial/r_radial/  
LCMS/dt/delta_t/  
PRF/osl/r_out/  
PRF/radial/r_radial/  
PRF/dt/delta_t/  
LAB/dt/delta_t/
```

Run-level traceability

product_catalog.csv maps



and stores the canonical species of each primitive channel.

Global vs. sliced output

- ▶ OSL fit results are written **only** for the **global sample**.
- ▶ Kinematic slices contain radial and signed Δt results.
- ▶ Slice names use the actual configured edges.

Examples:

mT_slice0.5-0.7

kT_slice1.0.35-0.6__mT_slice1.0.75-1

Clean output boundary

The production root must be **absent** or **empty** before SAFA writes the results, preventing stale files from being mistaken for current output.

10. Output Files and Fit Diagnostics

Spatial distributions

distribution.csv

- ▶ bin edges and center;
- ▶ normalized density p_i ;
- ▶ counting error δp_i ;
- ▶ valid Gaussian and/or mixed fitted PDFs for each enabled estimator.

fit_parameters.csv

- ▶ model, estimator and parameter value;
- ▶ asymmetric MINOS errors σ_- , σ_+ ;
- ▶ fit validity and explicit failure reason;
- ▶ $q_{\min}$, selected region and N ;
- ▶ MIGRAD, covariance and MINOS diagnostics;
- ▶ for the mixed fit: diagnostics for all 36 deterministic starts, terminal $(R_{\text{core}}, R_{\text{tail}}, f_{\text{core}})$, and selected-basin multiplicity.

Signed Δt output

distribution.csv

$$\Delta t_i, \quad p_i, \quad \delta p_i$$

statistics.csv

$$N, \quad \langle \Delta t \rangle, \quad \sigma_{\Delta t}, \quad \delta \sigma_{\Delta t}$$

There is no fit_parameters.csv for Δt because these distributions are summarized by moments rather than by the Gaussian / mixed fit machinery.

Auditable fit output

SAFA stores not only the published fit parameters, but also the numerical information needed to trace how the selected solution was obtained.

Diagnostic run summary

The executable also reports to stdout particle/pair rejections, formed and valid pair counts, P/PR/PRD routing, slice counts, and histogram-range warnings.

Output stage rule

The production writer performs no new scientific calculation: it serializes the already-derived post-sample analysis state.

Full Pipeline Overview

